OpenWebSearch.eu - Building an Open Index of the Web on HPC Infrastructure

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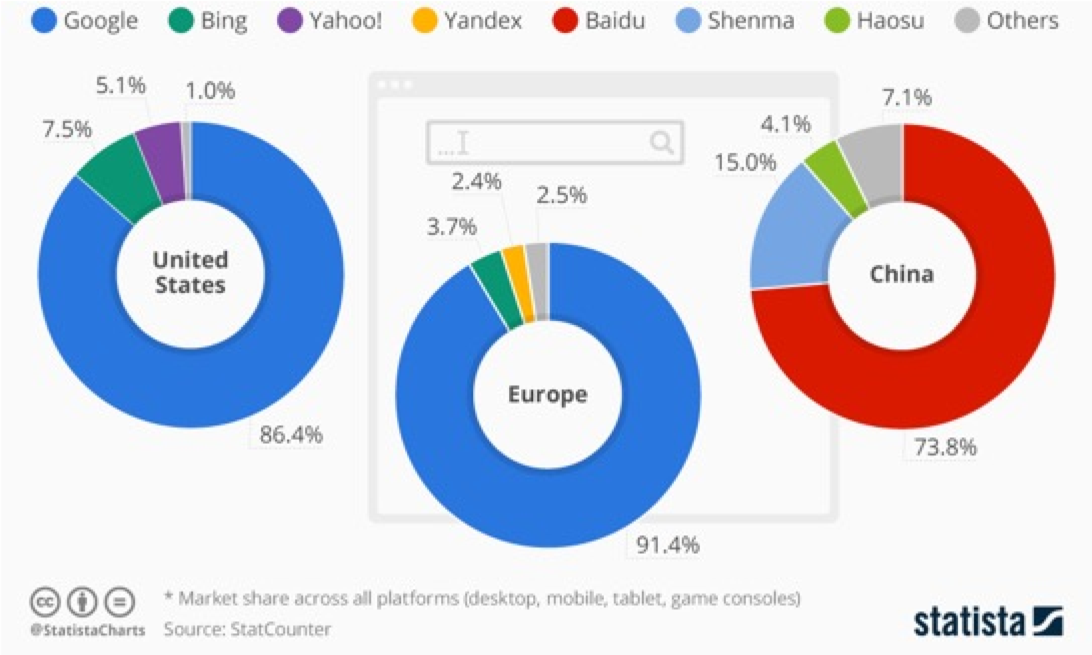
Web Search Today

A critical resource managed as an oligopoly

Two properties that don't fit

- A **critical infrastructure** for society, comparable to satellite navigation
- A **market oligopoly**: dominated by a small number of gatekeepers (Google, Microsoft, Baidu, ...)



But does it matter? Yes — on four dimensions

User Choice & Democracy

- Single gatekeeper for society's information
- Search rankings can shift voting preferences by **20–72%** (SEME)
- Monopolistic ranking = single point of failure for information quality

Societal & Security Risks

- European OSINT depends on US-controlled infrastructure
- Filter bubbles, algorithmic bias
- Misinformation & Disinformation

Untapped Richest Information Source

- 60 % of web pages carry structured data
- No open access to web-scale document collections
- Science, research and technology heavily depend on the Web as platform

AI & Innovation

- Web data is the fuel for LLMs, RAG, and agentic AI
- Closed indices block reproducible research & open innovation

> Being able to navigate the Web is as fundamental as GPS or the power grid

Challenges

THE WEB: VOLUME, VELOCITY, VARIETY

Careful: domains, sites, pages, and indexed documents are different units

VOLUME

1.15 Billion

Netcraft site responses

272.6 M domains · Dec 2024 Netcraft survey

Top 0.1%–1% of sites

may account for roughly half of all pages

heavy-tail intuition from large web graph studies

~ 400 Billion

documents in Google's search index

2020 figure cited in US v. Google testimony

Sites ≠ domains ≠ pages; use exact counts cautiously

VELOCITY

+8.6 M / month

monthly net change in Netcraft site responses

Dec 2024 versus Nov 2024 Netcraft survey

~ 3.2 / second

same monthly net change, averaged across December

149 ZB (global)

all digital data created / copied worldwide in 2024

IDC / Statista global datasphere estimate

The web changes constantly; crawling is continuous, not one-off

VARIETY

150+ languages

observed across websites

W3Techs usage survey

249 CMS Platforms

CMS platforms identified in the HTTP Archive dataset

HTTP Archive Web Almanac 2024 (Wappalyzer-based)

HTML · PDF · JSON-LD · media · APIs

content types to parse and process

HTML · PDFs · JS apps · schemas · media · APIs

Sources: Netcraft Web Server Survey Dec 2024 · large web graph studies (heavy-tail intuition) ·
US v. Google testimony on ~400B documents (2020 figure) · HTTP Archive Web Almanac 2024 · W3Techs · IDC/Statista

- High infrastructure costs
- Broad range of technical skills required
  - Big Data Infrastructure
  - Natural Language Processing, Image Analysis
  - Web Technologies
- Growing legal uncertainties

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High upfront costs to tap the Web as resources — especially small innovators and researchers are left behind

OpenWebSearch.eu — Our Mission

Building an Open Index of the Web and an open infrastructure to reduce upfront costs for researchers, innovators and companies in tapping the Web as a resource for search, web-analytics and AI.

Four Objectives

1. **Open Technology Stack** Open-source pipelines for crawling, preprocessing, indexing
2. **Resource Provision** Building a network of infrastructure providers (EuroHPC / AI Factories)
3. **Added Value Services** Dashboard, tools, data access APIs
4. **Bootstrapping the Ecosystem** Third-party calls, community building, applications and use-cases

14 Core Partners

Universität Passau

Webis.de

Universität Leipzig

Bauhaus-Universität Weimar

Radboud Universiteit

DLR

TU Graz

lrz

CERN

VSB Technical University of Ostrava

IT4Innovations National Supercomputing Center

CSC

ICT Solutions for Brilliant Minds

Infrastructure

open search foundation

nlnet FOUNDATION

suma-ev

A1

Research

NGOs

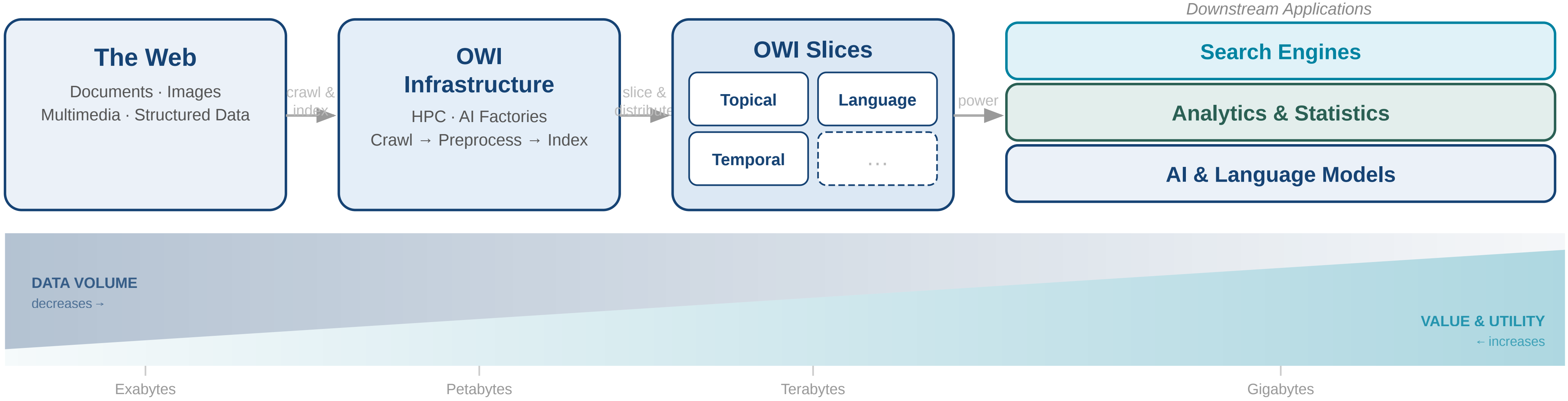
Businesses

Plus 16 financially supported third-party projects (FSTPss)

The Open Web Index (OWI)

A **Web Index** = a data structure for fast query-based access and ranking of web documents — the core of every web search engine

Our Proposition: A collaboratively created, open and transparent Web Index running on existing HPC computing centres / AI Factories in Europe



Granitzer, Michael, et al. "Impact and development of an Open Web Index for open web search." Journal of the Association for Information Science and Technology (2023).

The Open Web Index

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> **The Open Web Index** — a federated, daily-published web index built across European HPC sites and consumed as composable shards.

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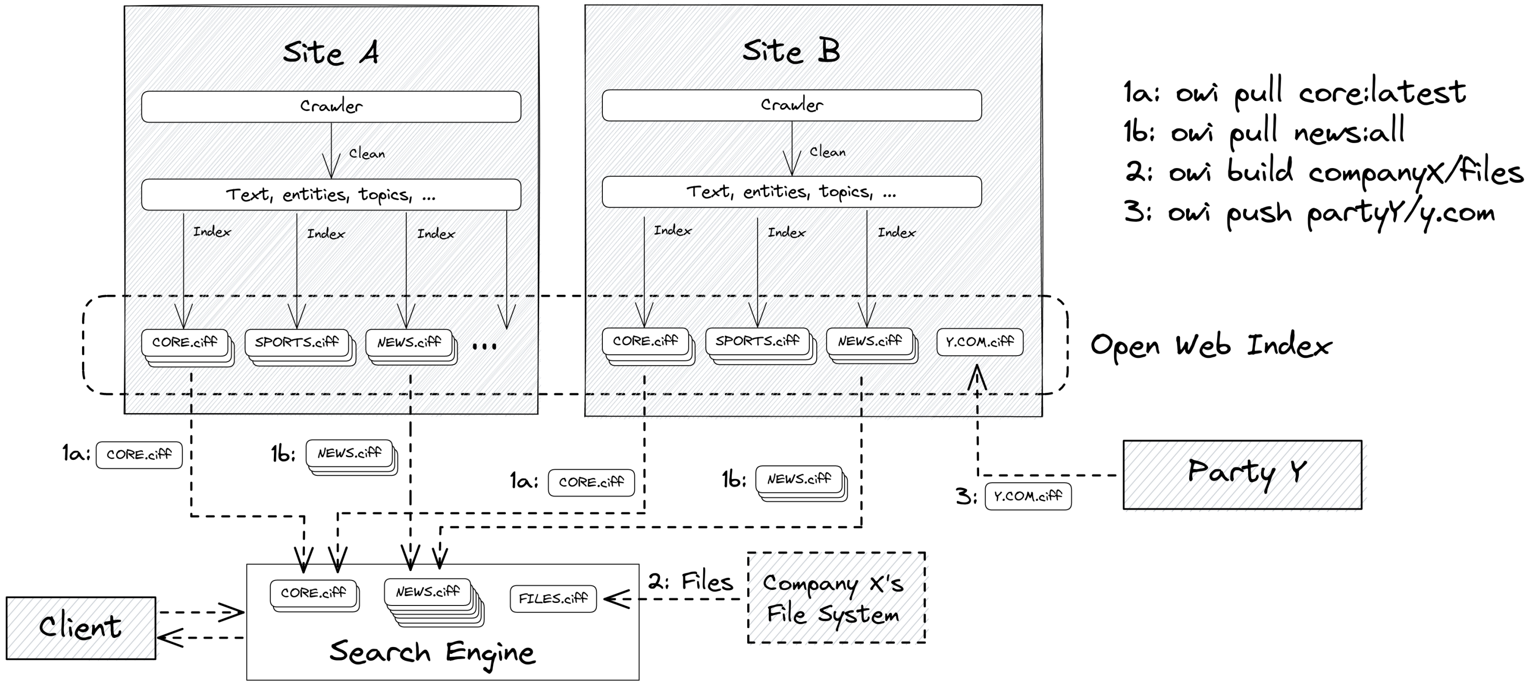


Federated production

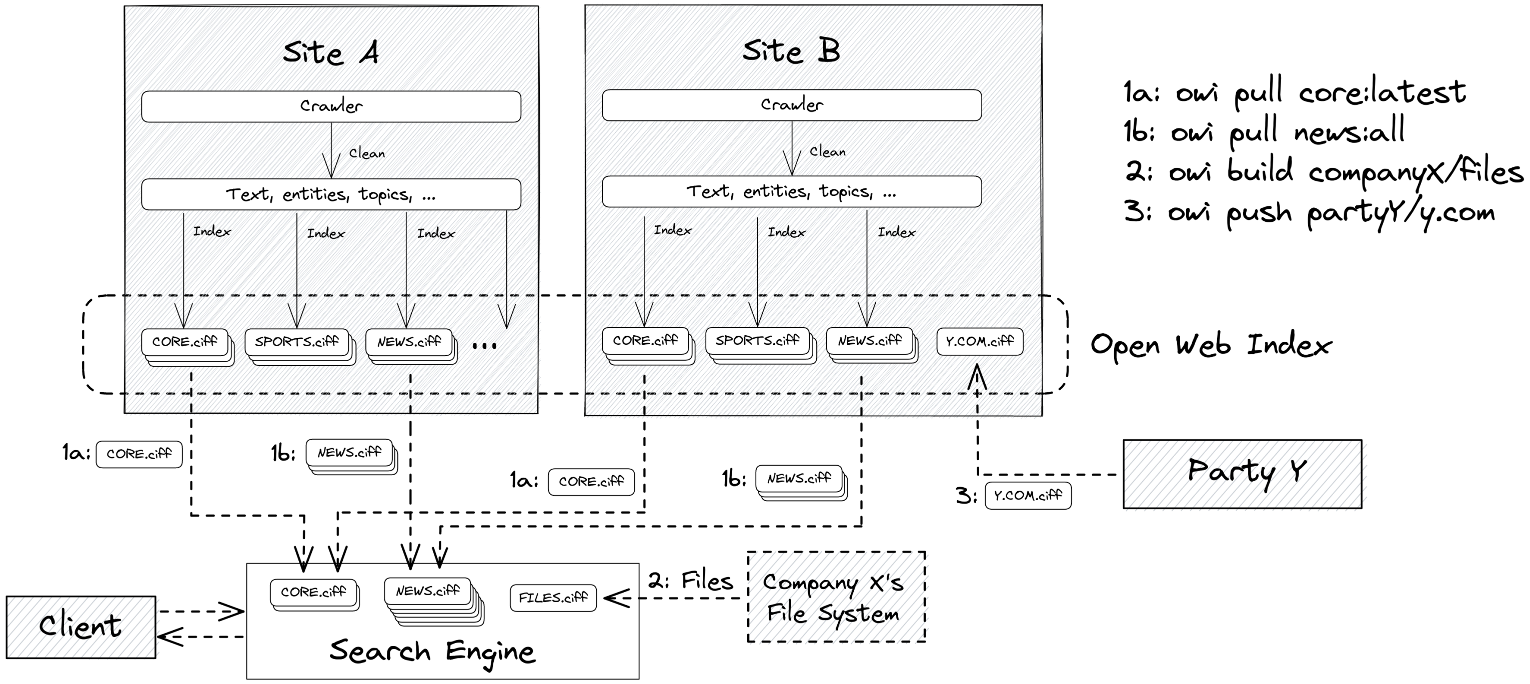
- each HPC site runs the same pipeline independently
- daily shards by date, language, topic

Three access verbs

- `owi pull` — download pre-built shards
- `owi build` — index your own data
- `owi push` — contribute shards back

The Open Web Index

The Open Web Index — a federated, daily-published web index built across European HPC sites and consumed as composable shards.



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Three output types per daily run — the building blocks of a shard:

OWI at a Glance — Statistics (Feb 2026)

Crawling

Pages crawled	~39 B total, ~12 B unique
Unique hosts	~500 M (extrapolated)
Crawling effort	1,927 machine-days
Avg. throughput	~20 M URLs / day

Index

Documents indexed	~10.7 B
Unique domains	~75 M
Public datasets	511 main / 1,511 total
Index size	45 TB (main) / 50 TB total
Embeddings	119 M docs · 884 GB

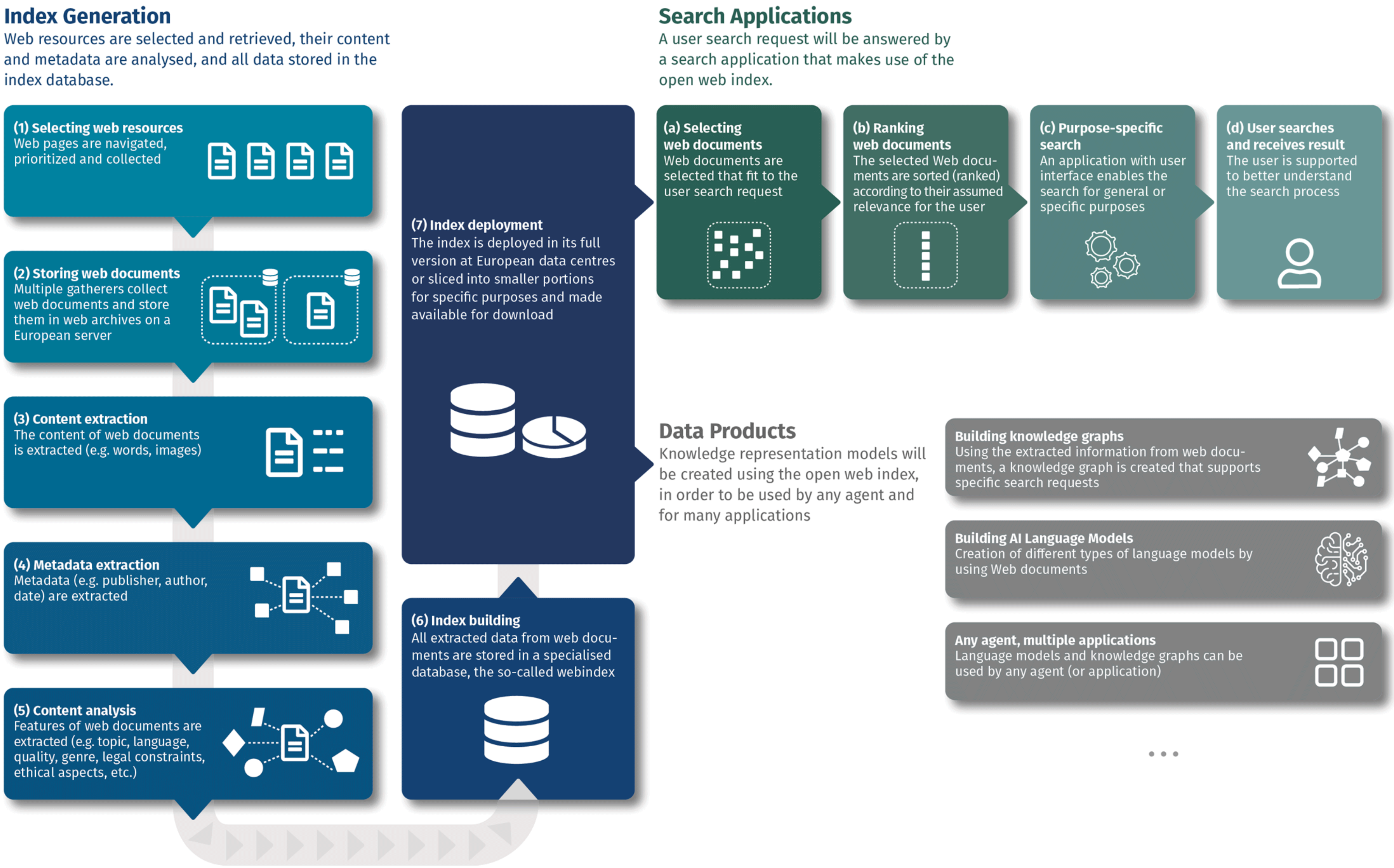
Top languages: English (38.9%), German (7.2%), Spanish (6.4%), French (5.7%), Russian (5.2%)

Multimedia Objects: Roughly 2-3 quality images and 0.5 Videos per Web-Document

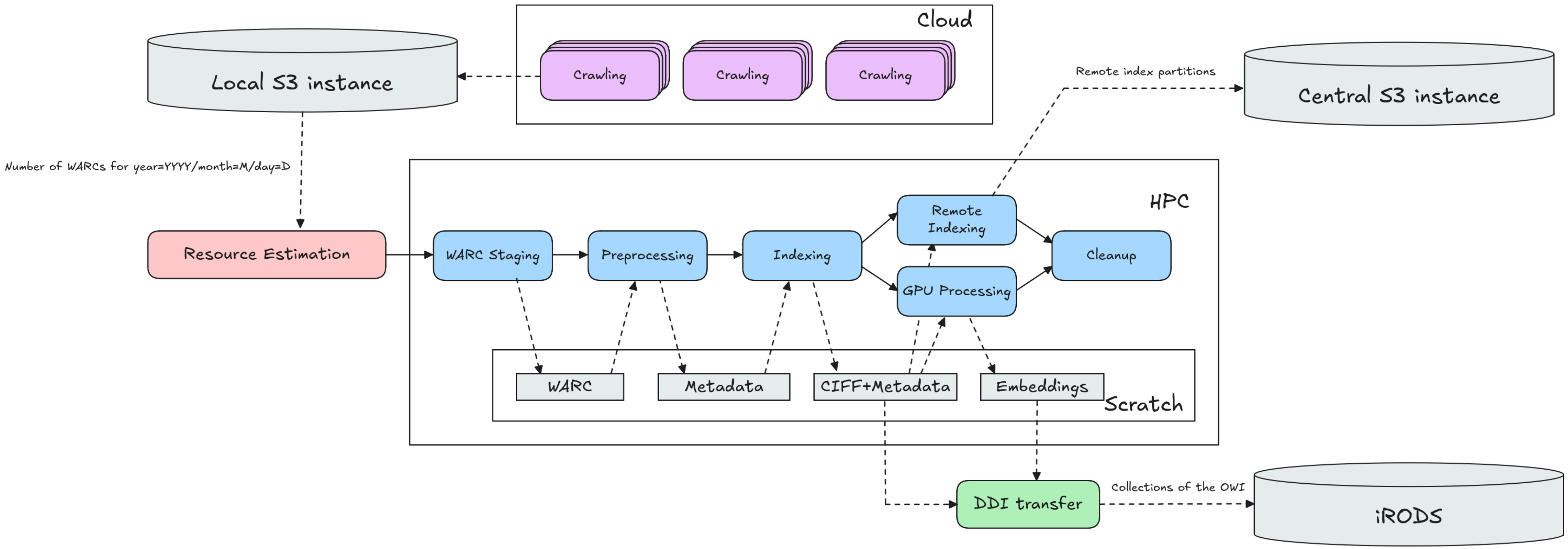
Comparison: To meet commercial index size, OWI would need to scale by a factor of 20× — factor 10–50× for multimedia

Building the Open Web Index on HPC Infrastructure

Conceptual Workflow



Daily Workflow Pipeline



Crawlig Infrastructure (2026)

- **25 VMs** across IT4I, LRZ, CSC, CERN
- **100 M URLs / day** crawling capacity
- **~1.1 PB** raw data · **~50 TB** index · **810+ public datasets**
- Federated storage: iRODS + S3, orchestrated via LEXIS

Limitations as of 2026

- TLR 5/6: it is not a product (yet)
- No 24/7 Operations
- Limited funding - minimum engineering / improvement

The OWI as HPC Workload

A heterogeneous, recurring, web-scale workflow: each stage shifts the bottleneck, and the whole pipeline runs daily across federated sites. That combination only fits HPC.

Stage	Dominant pressure	Trend
Crawling	network, scheduling, politeness	stable
WARC processing	I/O, parsing, decompression	stable
Indexing	memory, sorting, CPU	stable
Embeddings	GPU throughput	↑ rising
Vector index	GPU memory, ANN compression	↑ rising

Why HPC, not commodity cloud

- network capacity for crawling at scale
- memory & CPU for daily index builds
- GPUs for semantic enrichment
- PB-class federated storage
- batch scheduling already in place at scientific centers

What is shifting

- GPU pressure rises as semantic enrichment becomes default
- vector indices must be scheduled next to classical indexing
- the HPC question is no longer only "can we index the web?" but "can we semantically enrich it often enough for AI search?"

No single commodity cluster covers all stages. HPC centers do — and turn the OWI into reusable public infrastructure rather than a private product.

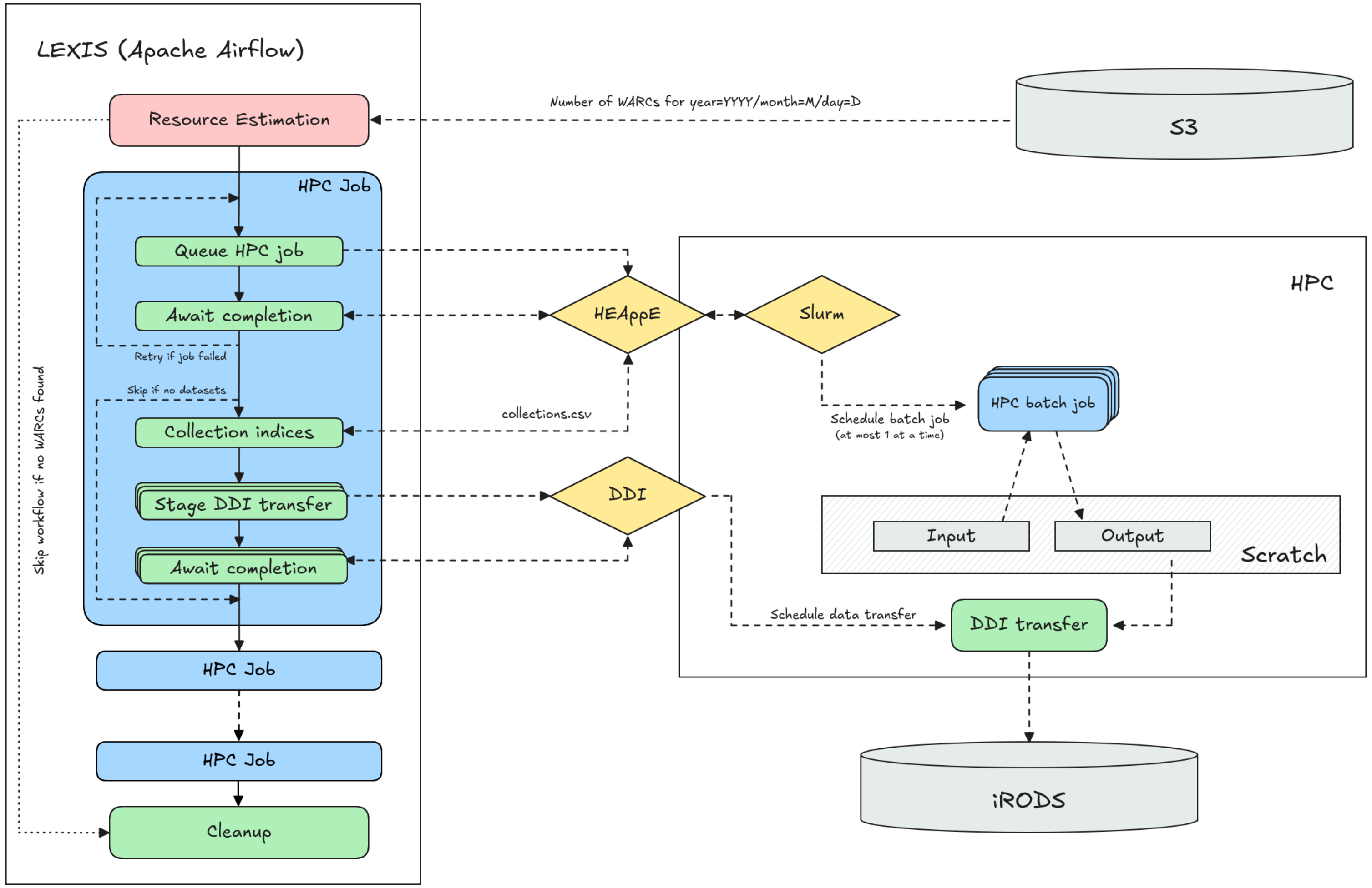
LEXIS: Orchestrating the Federation

The federation problem

- multiple HPC sites, each with its own scheduler, storage, identity
- the OWI must produce *aligned* daily output across all of them
- compute and data must meet — wherever the cycles happen to be

What LEXIS does

- coordinates OWI workflows across participating HPC / data centers
- stages input data and publishes daily OWI outputs
- bridges federated storage and compute jobs
- makes the build process repeatable

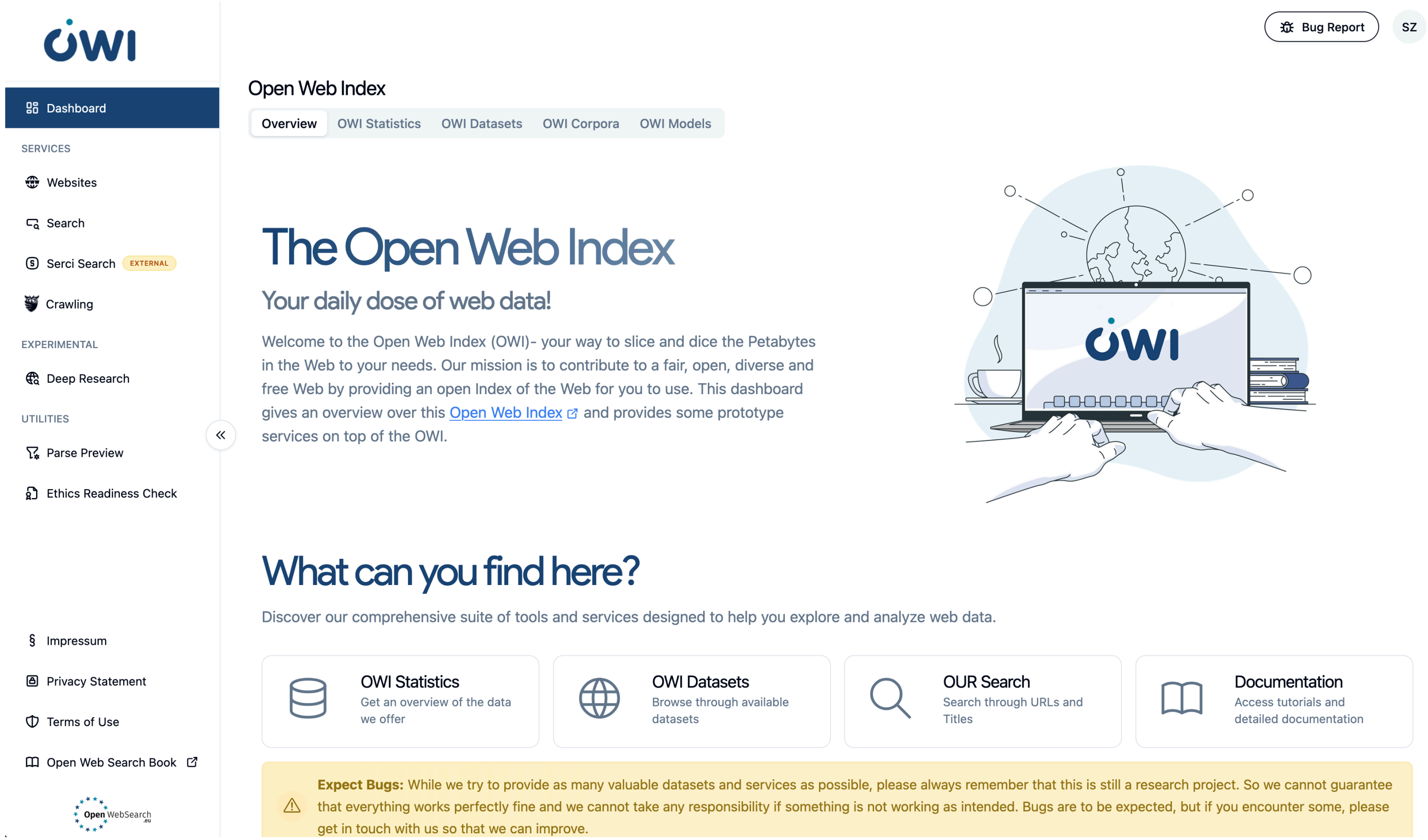


Workflow orchestration via LEXIS

> LEXIS turns distributed HPC resources into a repeatable, federated Open Web Index build process — including data distribution across centers.

Accessing the Open Web Index

The Dashboard — openwebindex.eu

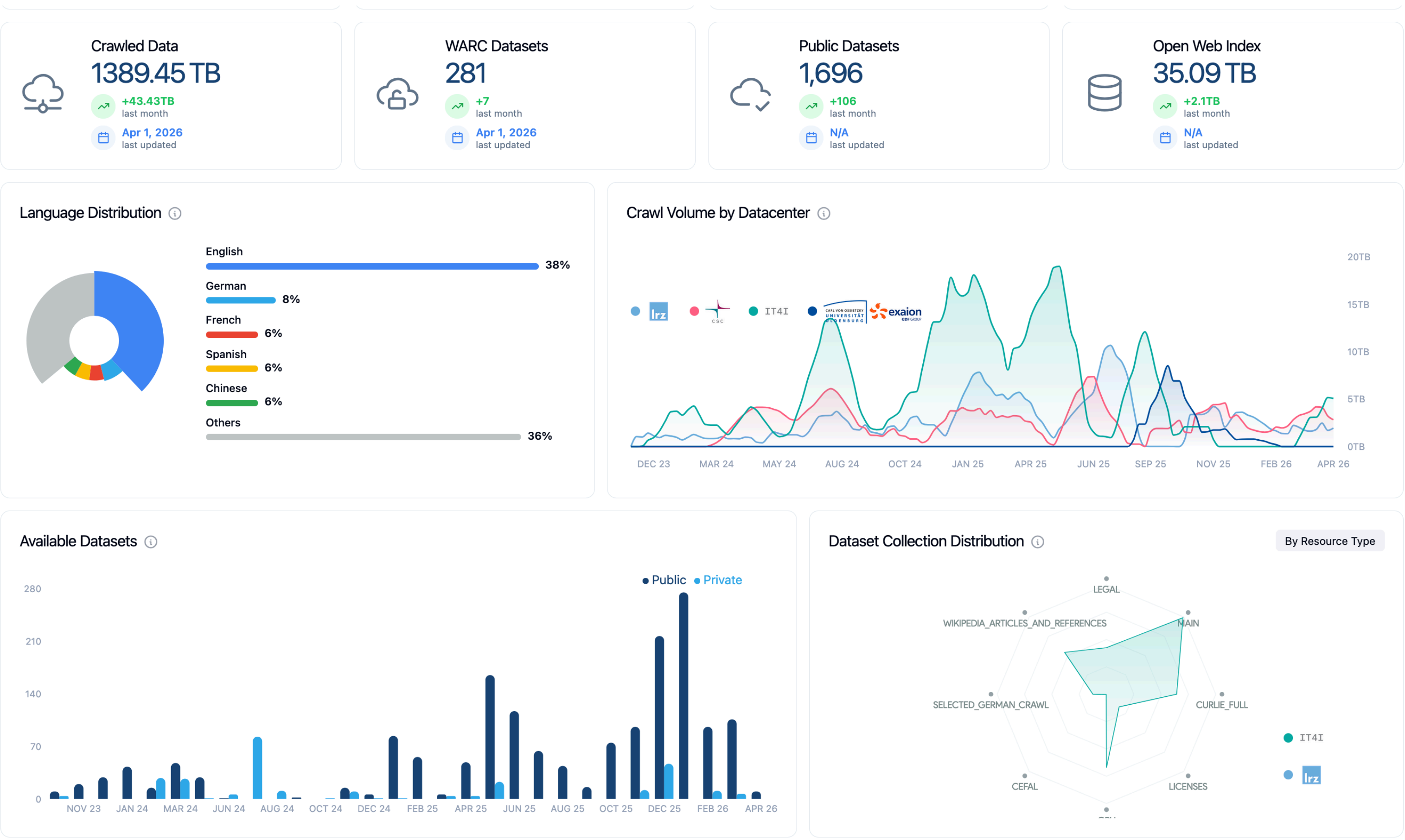


424 registered users · 17,000+ unique visitors · 70,000+ page views

Central entry point for the Open Web Index openwebindex.eu

- **OWI Statistics** — crawl volume, index size, language distribution, dataset timeline
- **OWI Datasets** — browse, filter and download index shards (by DC, date, format, language)
- **Search** — demo content search + SERCI integration

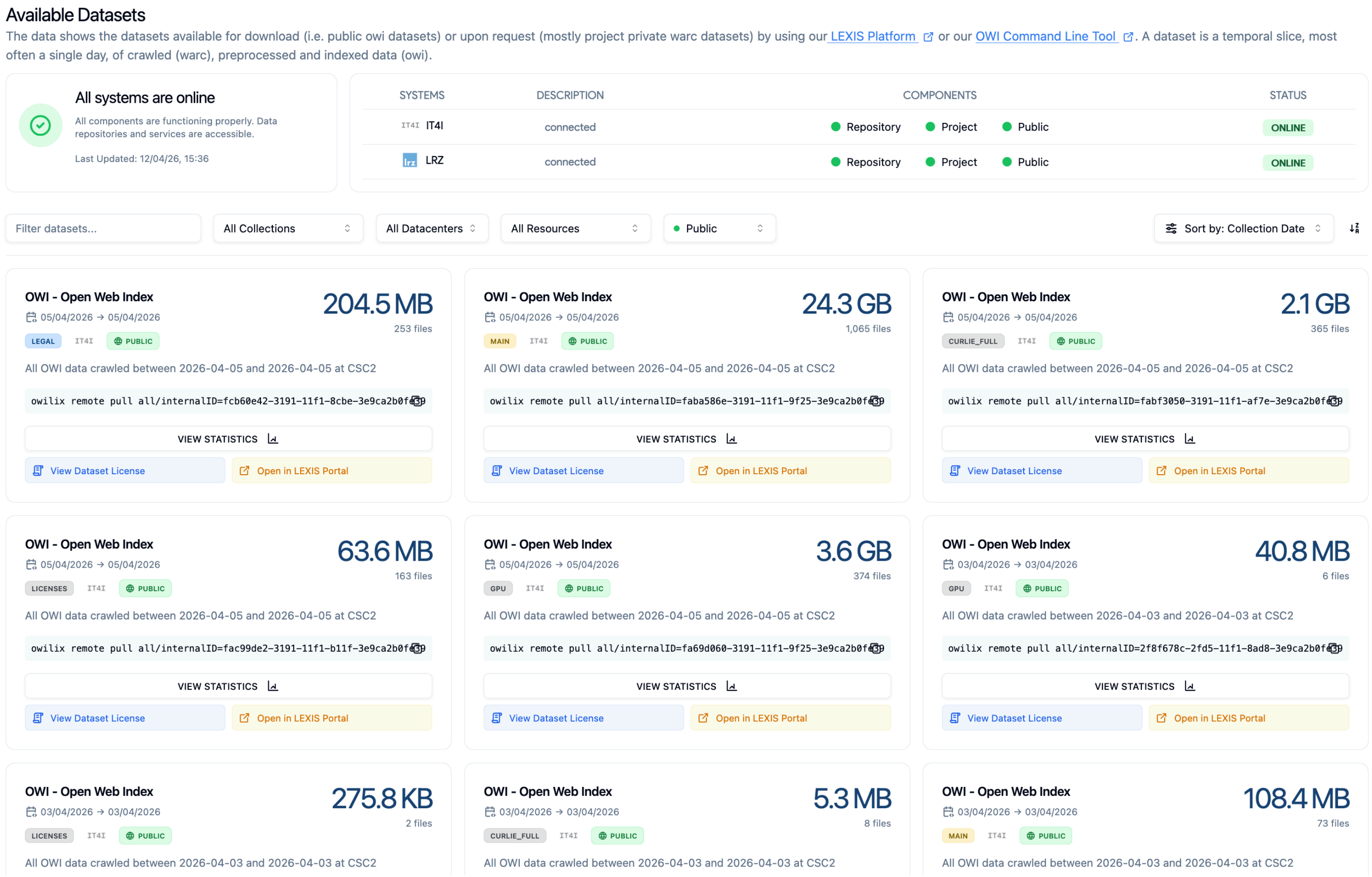
Dashboard — Statistics & Datasets



OWI Statistics (04/26)

- **1,389 TB** crawled · **281** WARC datasets · **1,696** public datasets · **35.09 TB** index
- Language distribution chart (English 38%, German 8%, French 6%, ...)
- Crawl volume over time per data center + dataset collection distribution by type
- Datasets over time

Dashboard — Statistics & Datasets



OWI Datasets Tab

- Status of repositories
- Filter by datacenter, format (WARC / Parquet / CIFF / Embeddings), language, date
- Dataset cards with download links — LEXIS Portal or owilix CLI
- Parquet file preview directly in the browser

Download Tool: owilix

owilix — The OWI Command-Line Tool

Git-like versioning for index datasets — pull, build, push:

```
owilix remote pull all:latest/collectionName=main
owilix local ls all:latest
```

Slice by: **domain**, **language**, **topic** (Curlie), **sitelist**, **structured data**, **outlinks**

Authentication via B2ACCESS / Keycloak (LEXIS SSO)

OWILIX - Open Web Index CLI

version 5.3.1 python 3.11+ license Apache 2.0

A command-line tool for managing OpenWebIndex datasets across local and remote repositories.

Quick Start

One-Line Install (Recommended)

```
curl -fsSL https://openwebindex.net/owilix/install.sh | sh
```

https://opencode.it4i.eu/openwebsearcheu-public/owi-cli/

```
~ owilix remote search "michael granitzer" --limit 10
Searching remote index: terms=['michael', 'granitzer'] language=eng representation=main_content mode=OR limit=10
Found 10 results in 113.4s
                    Search results (10 results in 113.4s)
```

score	date	url
3.088	2025-12-30	https://easychair.org/smart-program/ICADL2024/person16.html
3.010	2026-02-19	https://www.wolframalpha.com/input/?i=michael
3.010	2026-01-19	https://www.michaelmoennich.com/
3.010	2025-12-08	https://michaelkoenig.de/
3.010	2026-01-05	http://www.michaelmoerk.dk/
3.010	2026-01-22	https://michaelkoenig.de/
3.010	2026-02-18	https://www.sktthemes.org/forums/users/dedicationpt/replies/
3.010	2026-02-26	https://www.abeldiaz3.com/tag/michael/
3.010	2026-02-20	https://discourse.psychopy.org/u/michael
3.010	2026-02-27	https://thedigitalgobag.com/author/michael/

Remote Index — Query first, then Download

- Parquet partitions directly queryable via **DuckDB** or **Apache Spark**
- Full-text search
- SQL access: filter by language, domain, date — no full-shard download needed
- Enables search + filtering + download (but still in alpha phase)

The OpenWebSearch Book

Comprehensive Documentation at openwebindex.eu/book

Covers the full OWI ecosystem:

- Getting started with owilix
- Building vertical search engines with MOSAIC
- Data formats: CIFF, Parquet, embeddings
- Pipeline architecture and HPC deployment
- Legal & ethical guidelines for index users

Published as an open, continuously updated online resource — aligned with deliverables and community feedback.

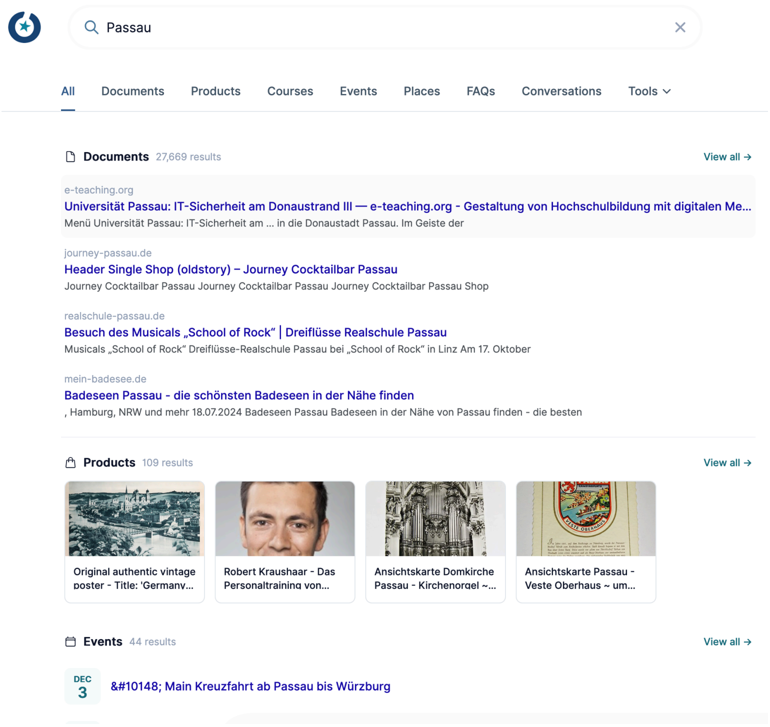
Access Modes at a Glance

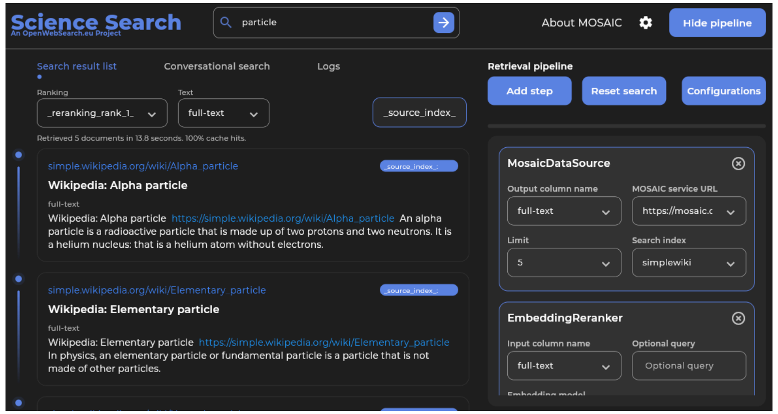
Tool	Mode	Best for
Dashboard	Browser	Explore, monitor
owilix	CLI	Download, slice, build
Remote index	DuckDB/SQL	Query + download
MosaicRAG	Conversational	RAG & chat over OWI

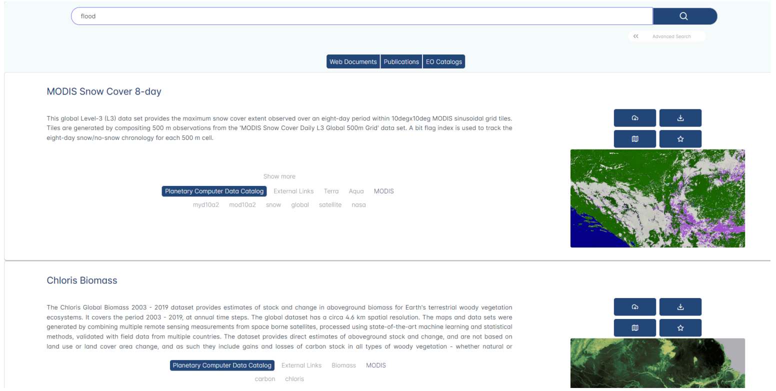
All software, data formats and access tools are **open source**

Using the Open Web Index

Using the Open Web Index — Search and RAG







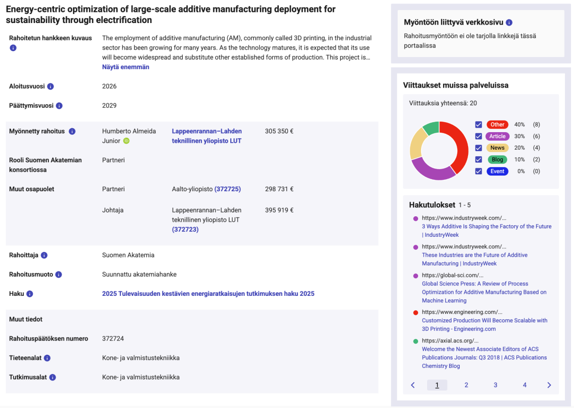
OUR²S
Hosted search UI and REST API for online retrieval directly over OWI content.

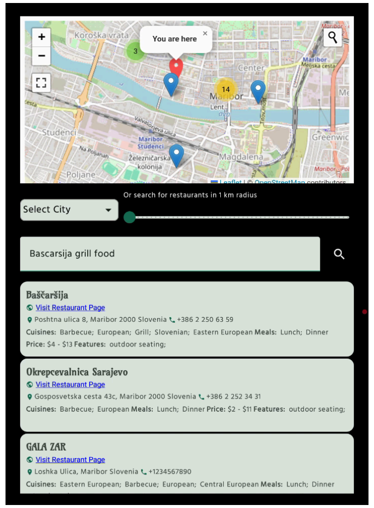
Mosaic vertical search
Domain-specific search engines for science, health, arts, and other focused collections.

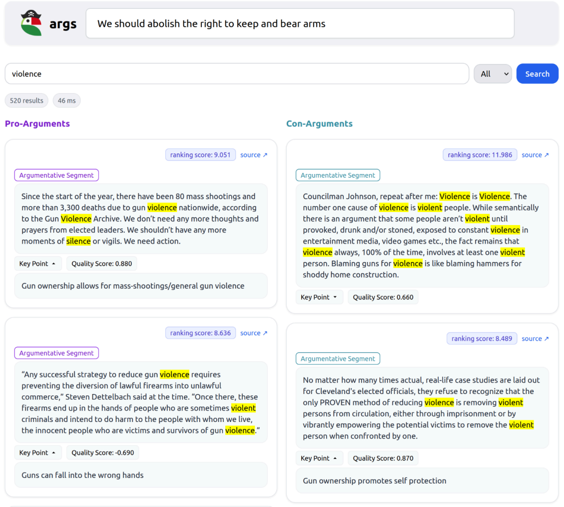
Earth observation search
RAG-based access to scientific papers, datasets, and geo-contextualized web content.

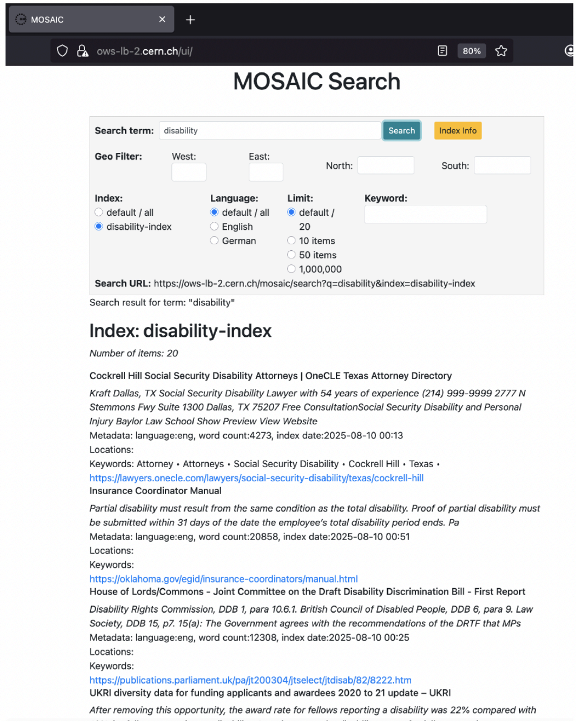
The OWI is useful when the application needs inspectable web-scale source material, not only a black-box answer.

Using the Open Web Index — Domain Applications









Research.fi
National research discovery enriched with OWI-derived scientific web content.

Restaurant recommender
Privacy-preserving local recommendation from extracted restaurant features.

Argumentation search
Pro/con argument retrieval over curated OWI slices.

Institutional search
CERN institutional search and disability knowledge search.

> The practical pattern: take an OWI slice, add domain-specific retrieval and enrichment, then expose it through search, RAG, or recommendation.

Using the Open Web Index - Data for AI and LLMs

Using the Open Web Index — Data & Analytics

Published Datasets

Dataset	Scale	Use case
WebFAQ	96M Q&A, 75 langs	QA training, cross-lingual IR
WebFAQ 2.0	198M pairs, 108 langs	Dense retrieval, hard negatives
Imprints	5.25M hosts, geocoded	Enterprise analytics, Eurostat
German Commons	154B tokens	German LLM pre-training
OWS-Curlie-2025	20.7M docs + qrels	IR evaluation benchmark

All datasets available at openwebindex.eu/corpora

Analytics Potential

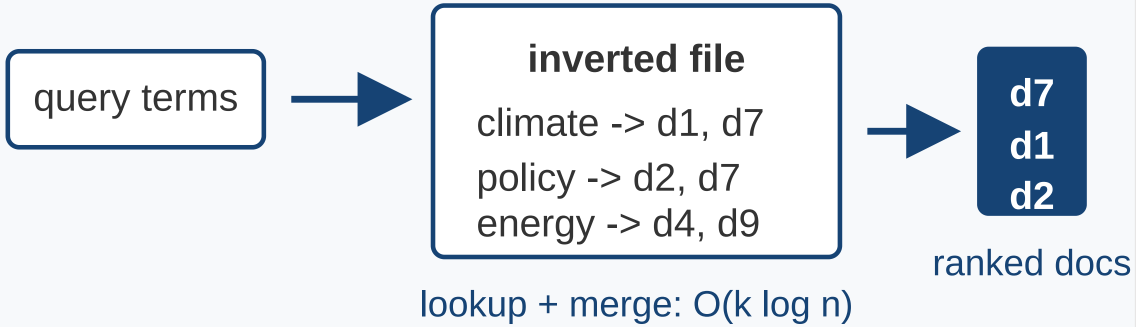
- **Official Statistics**: enterprise data extraction from legal imprints (Eurostat, Destatis)
- **Media Monitoring**: supply chain analysis, news monitoring (JRC)
- **AI Training Data**: domain-specific corpora for LLM fine-tuning
- **Evaluation Benchmarks**: web-scale IR test collections with relevance assessments
- **Trend Analysis**: temporal web monitoring at Petabyte scale

The OWI is not just for search — it is a data infrastructure for the web-scale analytics ecosystem

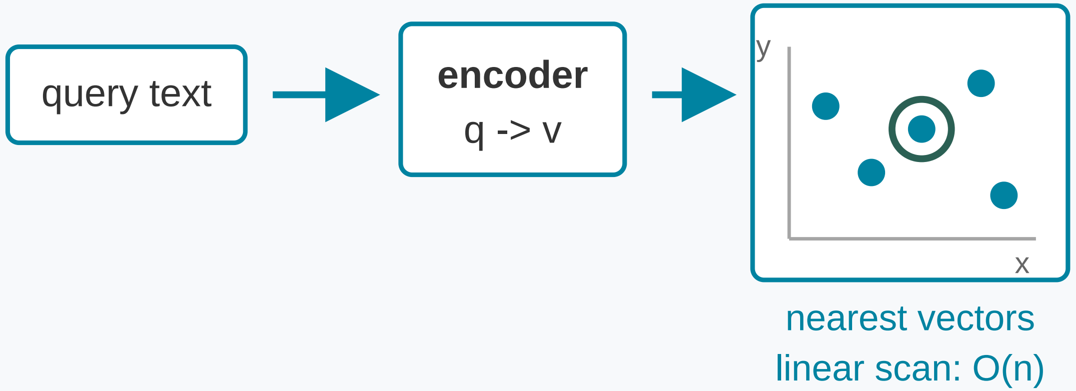
From Search to AI-Based Search

Retrieval Modes on the Open Web Index

Sparse index



Dense index



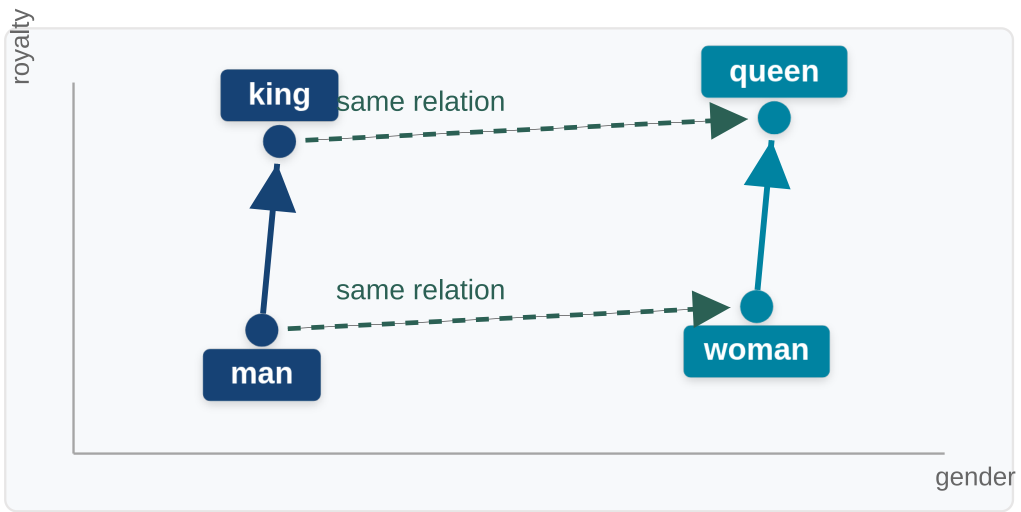
Sparse search
- exact terms and fields
- BM25-style ranking
- efficient inverted indices
- strong for names, facts, rare terms

Dense search
- semantic similarity in vector space
- robust to paraphrases
- useful for RAG and question answering
- needs vector storage and fast ANN

Hybrid search
- combine lexical and semantic evidence
- use metadata filters
- rerank with embeddings or LLMs
- balance precision and recall

The IR progression: lexical matching gives efficient candidates; embeddings add semantic matching; hybrid search combines both and raises the serving question.

Embeddings Encode Relations



vector(king) - vector(man) + vector(woman) ≈ vector(queen)

Analogy example after Mikolov et al., *Efficient Estimation of Word Representations in Vector Space* (Word2Vec), arXiv:1301.3781.

The classical word embedding example illustrates the intuition:

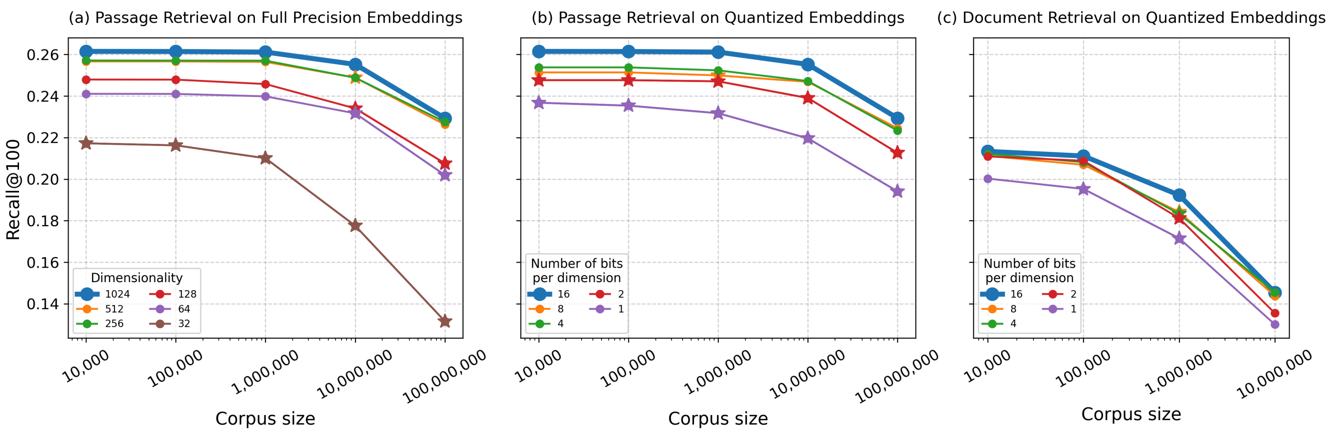
```
king - man + woman ≈ queen
```

Embeddings place objects in a vector space where some semantic relations become approximately geometric operations.

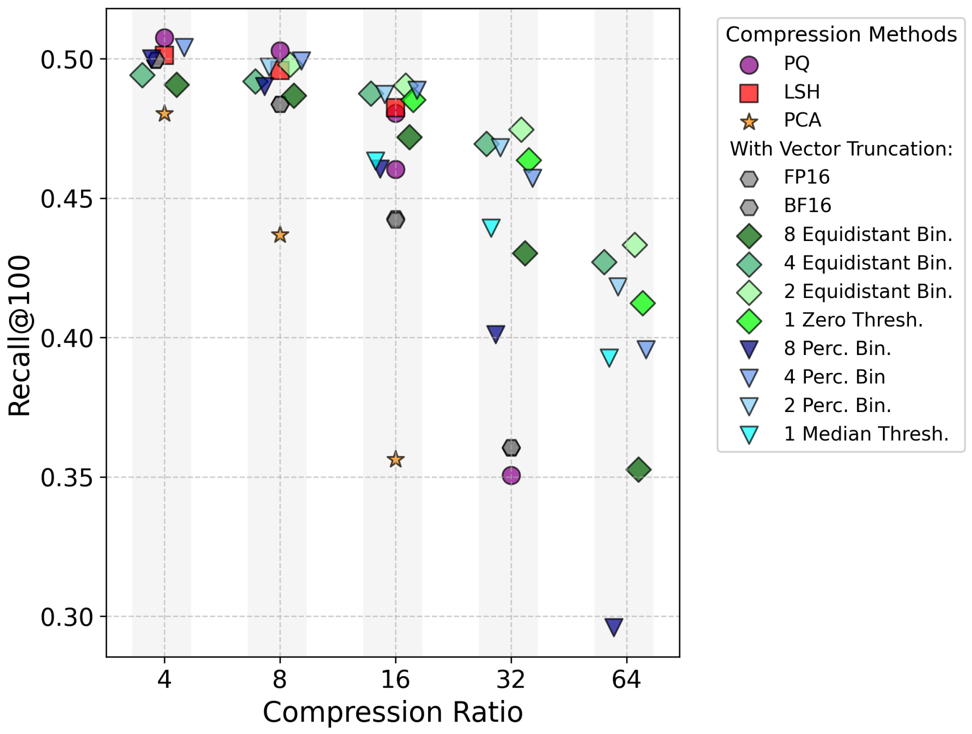
For search this creates a systems question: low-latency vector retrieval needs large vector indexes, often RAM-resident or close to RAM, plus substantial persistent storage.

Order of magnitude: at 1024-dim float32, 1 M vectors ≈ **4 GB** and 100 M vectors ≈ **400 GB**. The OWI currently ships **119 M dense vectors at ~884 GB**.

CoRECT: Evaluating Embedding Compression at Scale



Jina v3, CoRECT line chart, NDCG@10. Source: padas-lab-de.github.io/CoRECT



Jina v3, CoRECT Pareto plot, NDCG@10. Source: padas-lab-de.github.io/CoRECT

CoRECT

- Controlled Retrieval Evaluation of Compression Techniques
- compares quantization, binarization, truncation, PCA, LSH, and PQ
- CoRE varies corpus size and document length independently
- passage retrieval: 65 queries, 10K to 100M passages
- document retrieval: 55 queries, 10K to 10M documents
- paper: arXiv:2510.19340

CoRECT: Compression Trade-Offs

Results

- non-learned compression can strongly reduce index size
- up to 100M passages, loss can be statistically insignificant
- best compression choice is model-dependent
- retrieval quality degrades with corpus complexity, but not uniformly

Serving conclusion

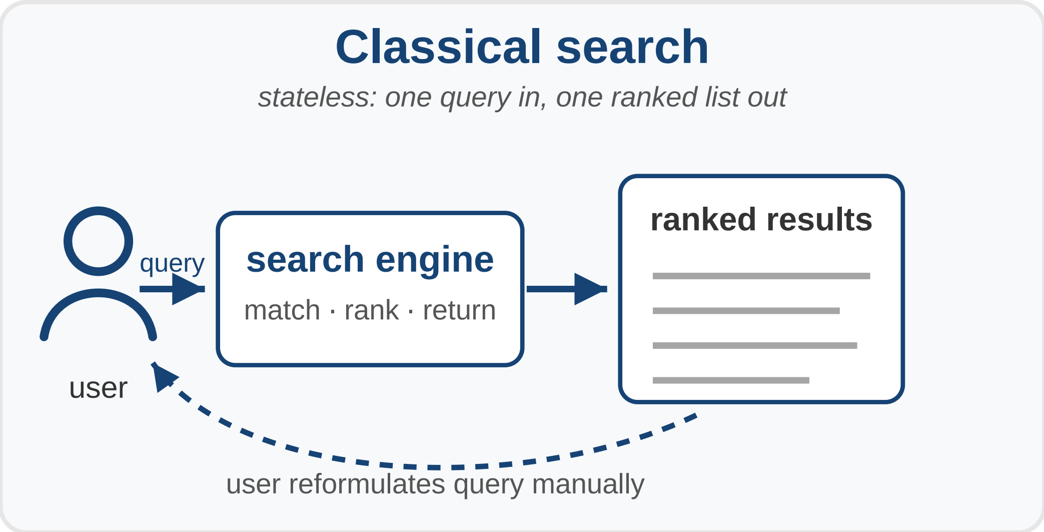
- uncompressed dense indexes are expensive to keep fast
- FP casting, scalar quantization, truncation, binarization, LSH, PCA, and PQ occupy different cost-quality regions
- the best point is not universal; it depends on model, metric, and corpus complexity
- open theoretical question: which ranking families can binary vectors express under Hamming distance or binary dot product?

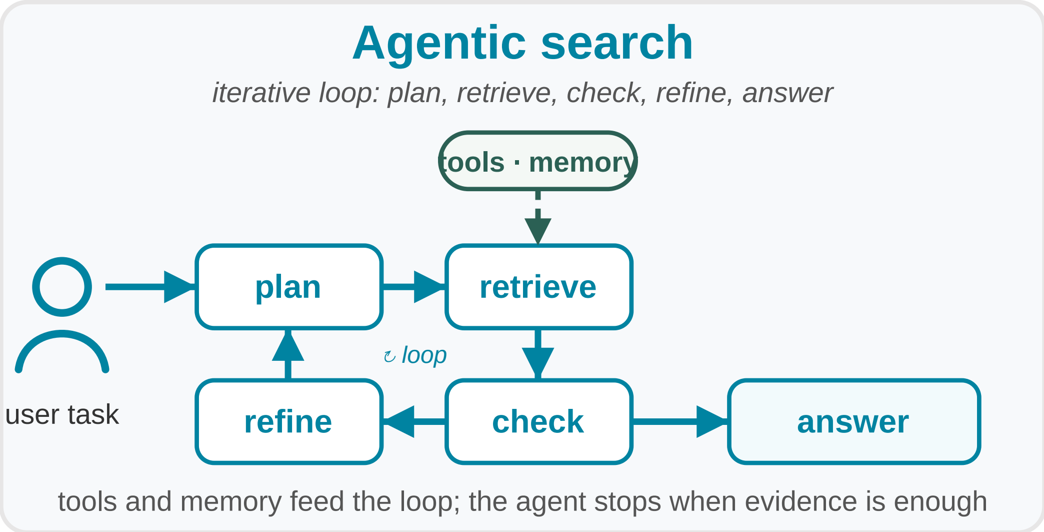
> Compression is a serving and storage decision: it trades RAM, storage, latency, and ranking quality.

≡

Agentic Search

From Search Engine to Search Agent





Classical search

- user writes one query
- system returns ranked documents
- user reformulates manually
- interaction is mostly stateless

Agentic search

- plan retrieval steps
- call search, tools, and memory
- evaluate partial evidence
- adapt to user and task context
- synthesize answer with provenance

> Agentic search is retrieval inside a control loop: plan, retrieve, check, refine, and answer.

PersonaRAG: Retrieval with User-Centric Agents

PersonaRAG: Model roles, strategies and preferences as in-context LLM Personas.

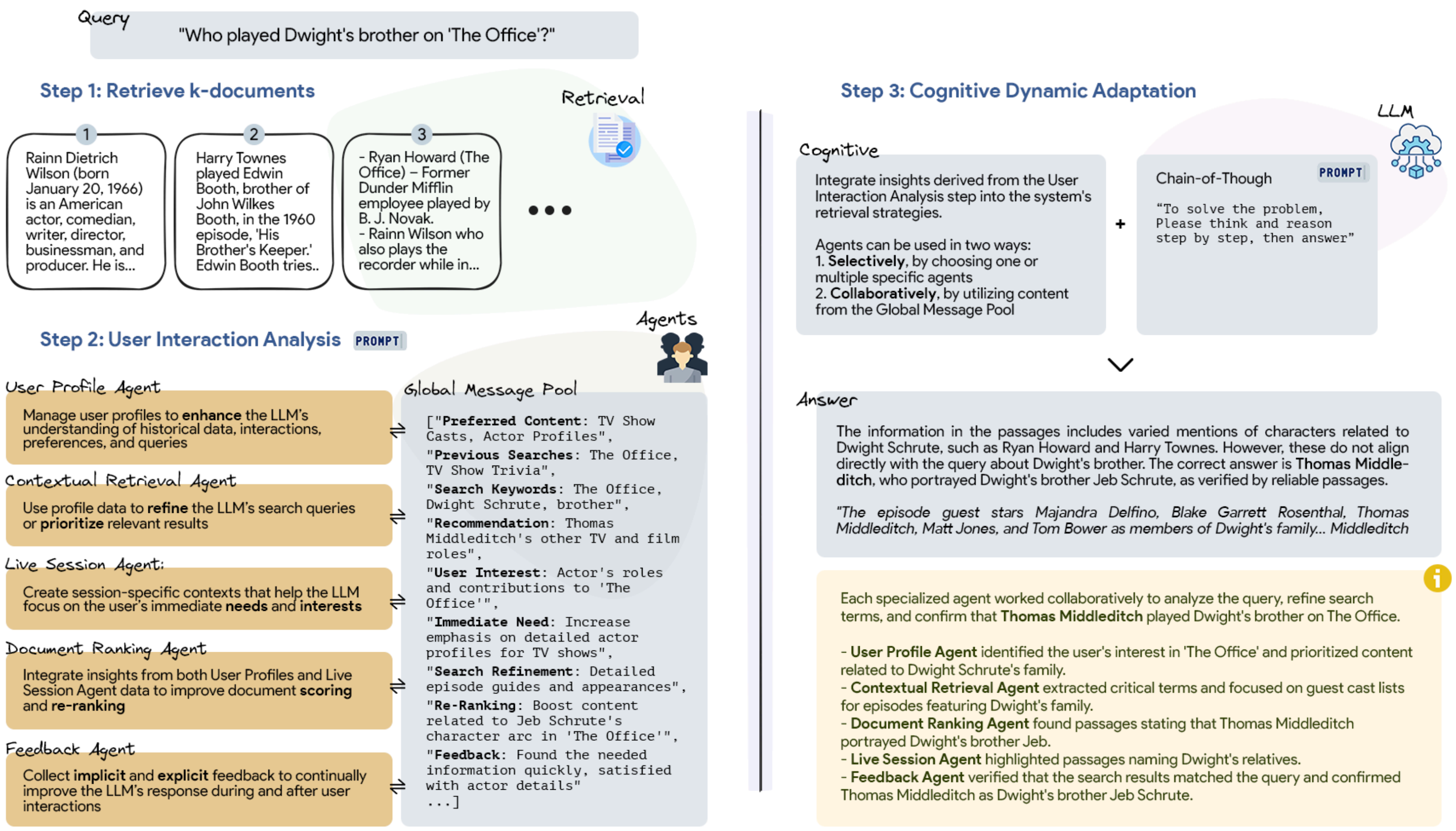


Figure from PersonaRAG, arXiv:2407.09394

Design tension

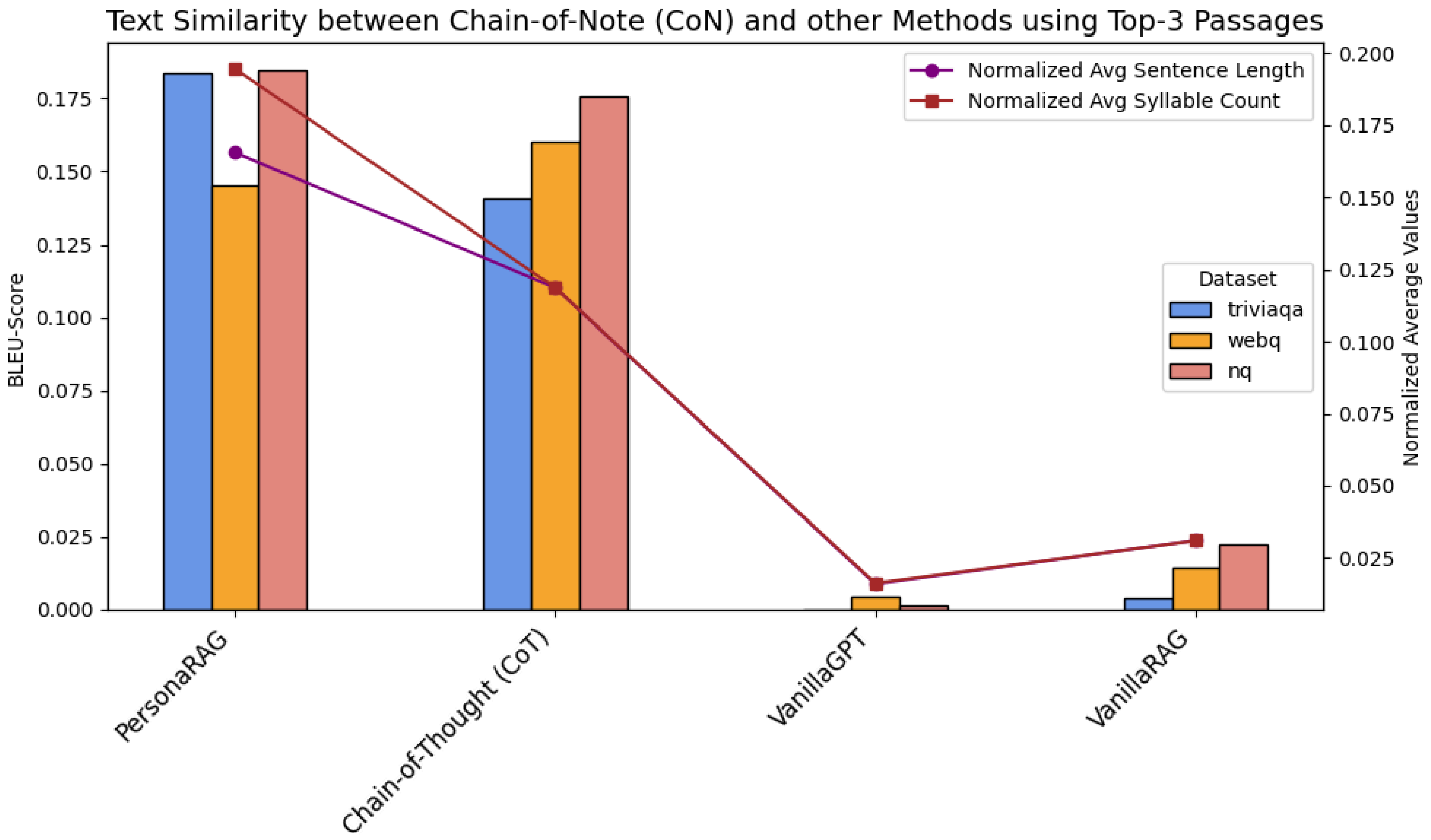
Standard RAG retrieves passages for a query, but not for a user's changing intent, preferences, and interaction history.

Mechanism

- user profile agent
- contextual retrieval agent
- live session agent
- document ranking agent
- feedback agent

> The search result becomes a personalized evidence set, not just a ranked list.

PersonaRAG: Evaluation and Findings



Text similarity experiment from PersonaRAG, arXiv:2407.09394

Conclusion for agentic search: personalization improves answers, but it also makes context management a first-class retrieval problem — what to keep, update, forget, explain, and expose to the agent.

How it was tested

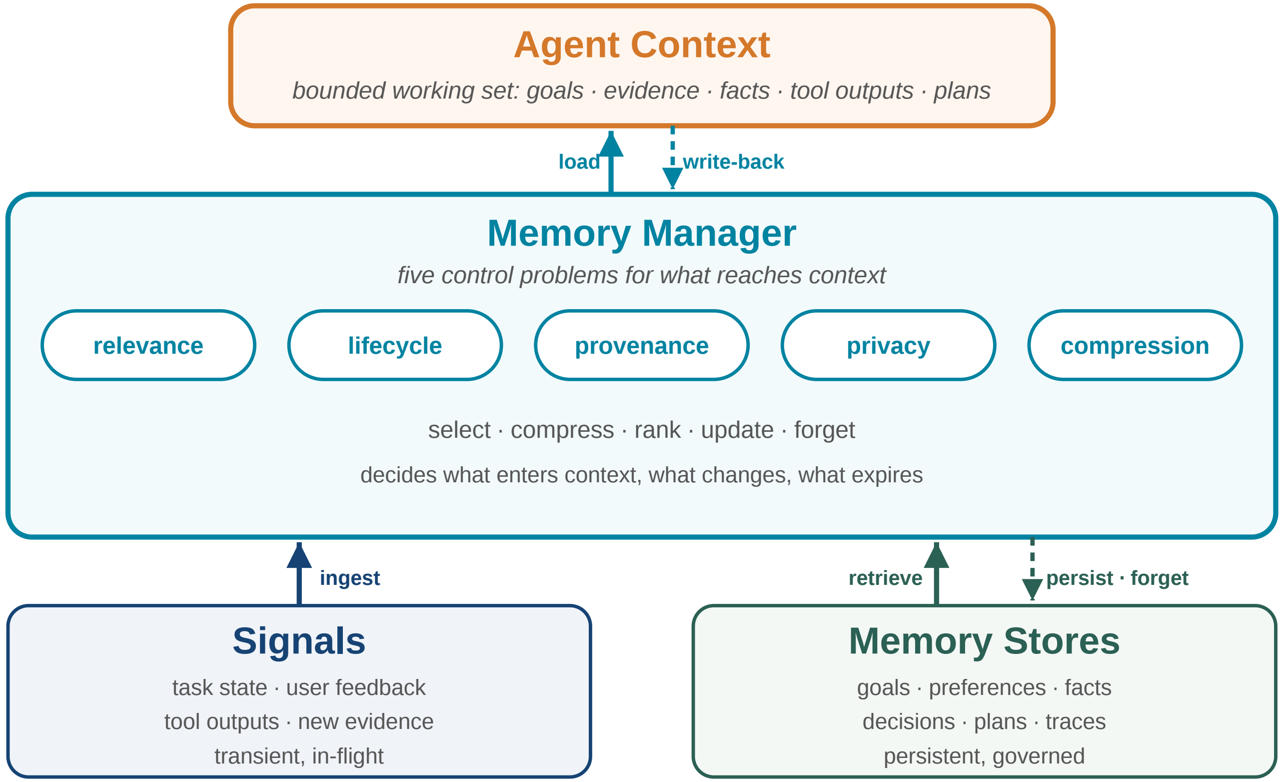
- NaturalQuestions, TriviaQA, WebQuestions
- 500 sampled questions per dataset
- GPT-3.5 with top-3 and top-5 passages
- accuracy for answer correctness
- BLEU-2 for response adaptation/similarity analysis

What changed

- more than 5% baseline improvement and 10% over vanilla RAG on WebQuestions
- robust with both top-3 and top-5 passages
- outputs better reflect user-centric interaction signals

Conceptual Agent Memory Components

Agentic search turns retrieval into an ongoing process: the agent must remember task state, user context, retrieved evidence, tool results, and unresolved questions. Due to attention degradation, **managing context** becomes key for agent performance



The memory layer becomes the boundary between raw retrieval and reliable agent behavior.

What memory stores

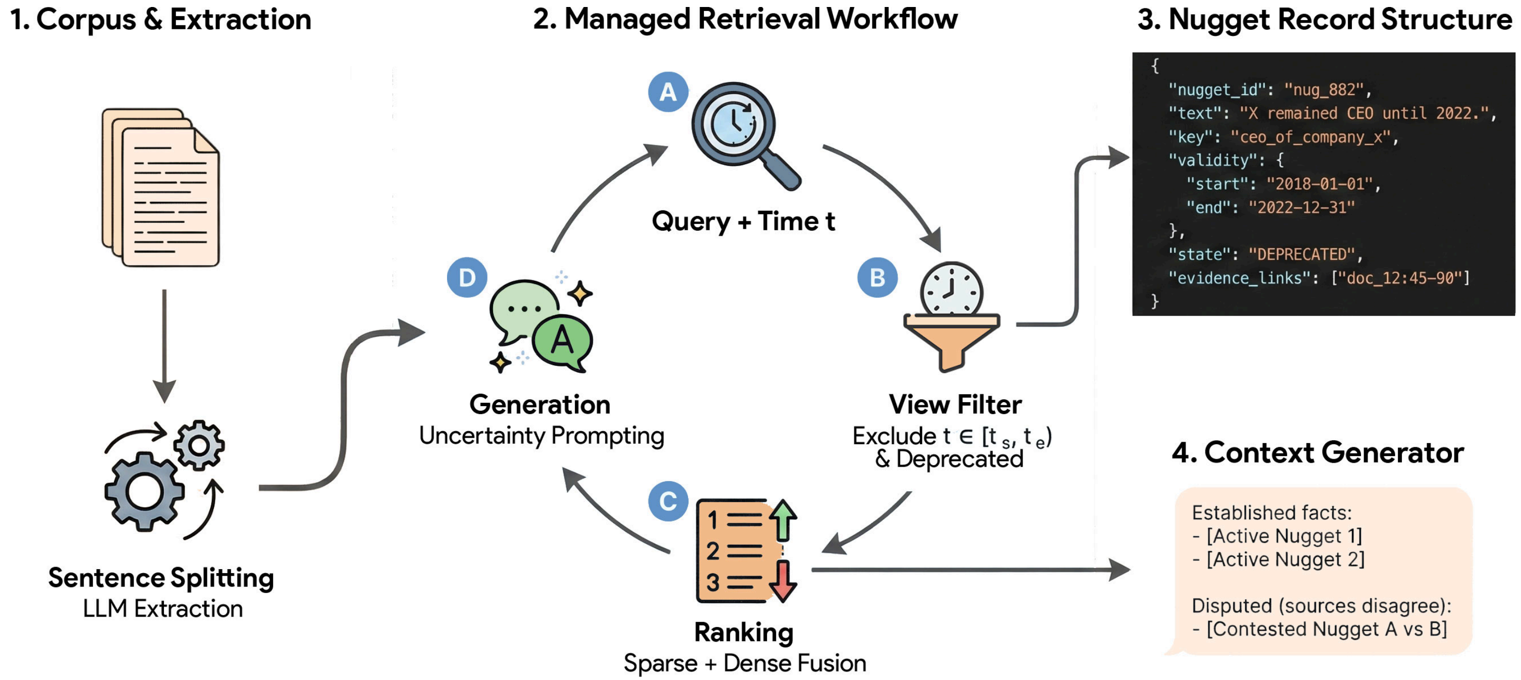
- session goals and constraints
- user preferences and feedback
- retrieved documents and snippets
- extracted facts and claims
- intermediate decisions and plans

What memory must control

- relevance: what should enter context?
- lifecycle: what changes or expires?
- provenance: where did a fact come from?
- privacy: what may be reused?
- compression: how much can fit?

NuggetIndex: Governed Local Fact Retrieval

Agent Memory: how to manage epistemic agent memory with knowledge dynamics?



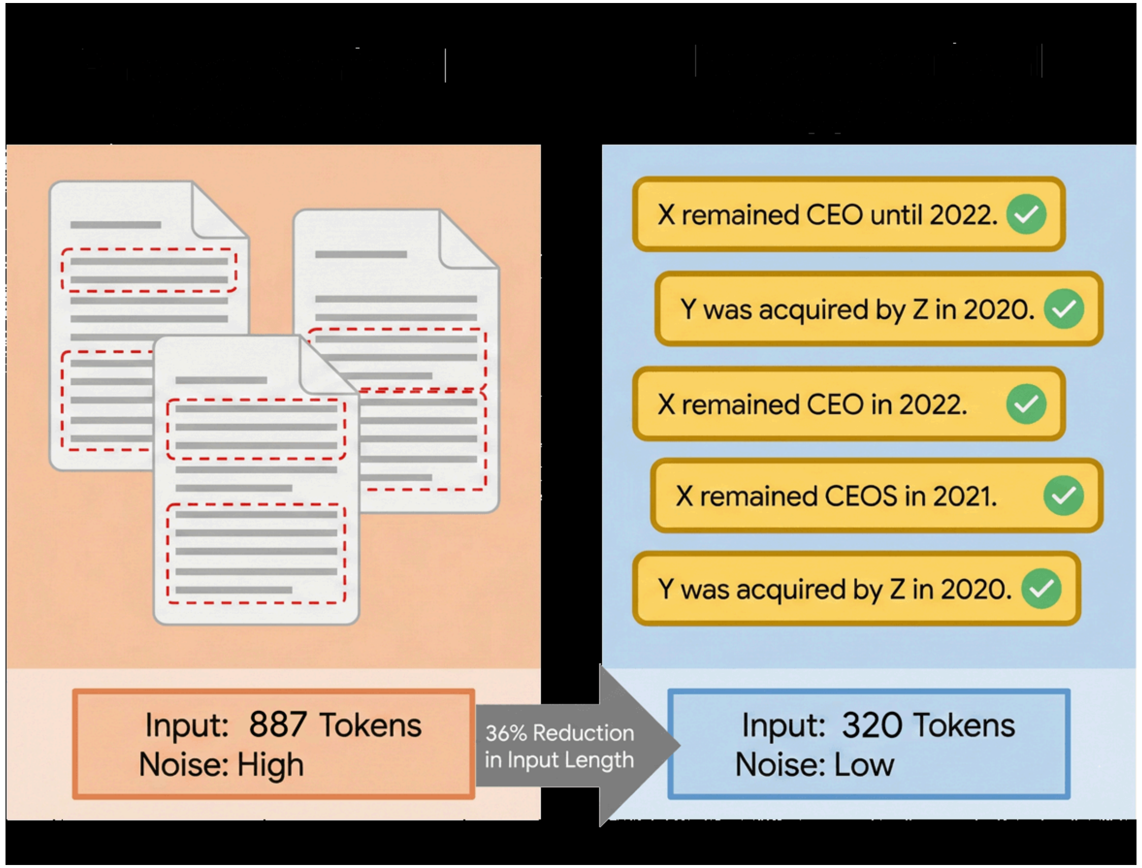
Architecture figure from NuggetIndex, arXiv:2604.27306

Design tension
Agents should not repeatedly retrieve long passages when the useful unit is an atomic fact with validity, evidence, and lifecycle state.

Mechanism
- extract atomic nuggets from text
- store validity intervals and evidence links
- assign active, deprecated, or contested state
- filter invalid facts before ranking
- retrieve compact local facts for generation

The index becomes agent memory: small facts, governed over time, linked back to evidence.

NuggetIndex: Evaluation and Findings



Efficiency figure from NuggetIndex, [arXiv:2604.27306]

How it was tested

- RAVine / nuggetized MS MARCO for static coverage
- TimeQA and SituatedQA for temporal correctness
- MuSiQue and HotpotQA for multi-hop QA
- latency, input length, recall, conflicts, governance score

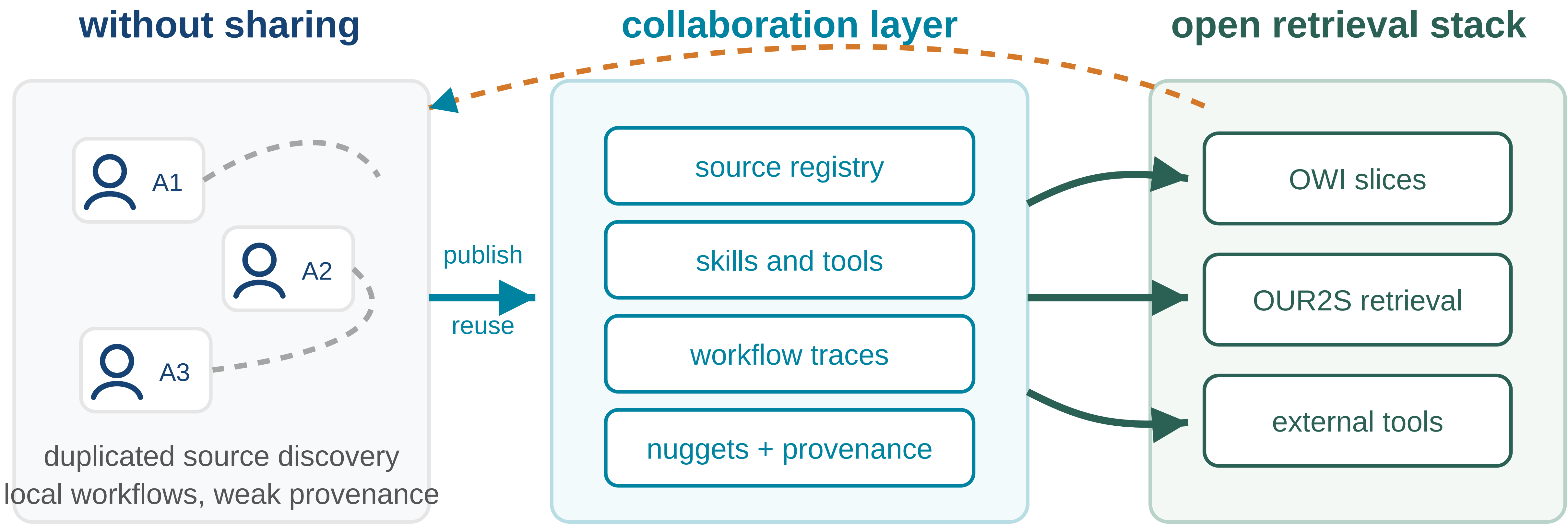
What changed

- nugget recall improves by 42%
- temporal correctness increases by 9.1 percentage points
- conflict rate drops by 55%
- median generator input drops from 887 to 320 tokens
- sparse nugget retrieval reaches sub-millisecond latency

Nuggetindex also works well with smaller reasoning models. This points toward smaller models for search systems and everyday tasks.

Agent Collaboration Systems

Why Agents Need Collaboration Infrastructure



shared context turns one-off agent searches into reusable infrastructure

Single-agent retrieval is limited

- each agent rediscovers sources
- useful context stays private to one run
- provenance is hard to share

Collaboration layer

- share memory across agents and sessions
- share useful sources and retrieval scopes
- expose skills and tools to other agents

- successful workflows remain local

- maintain reusable knowledge units

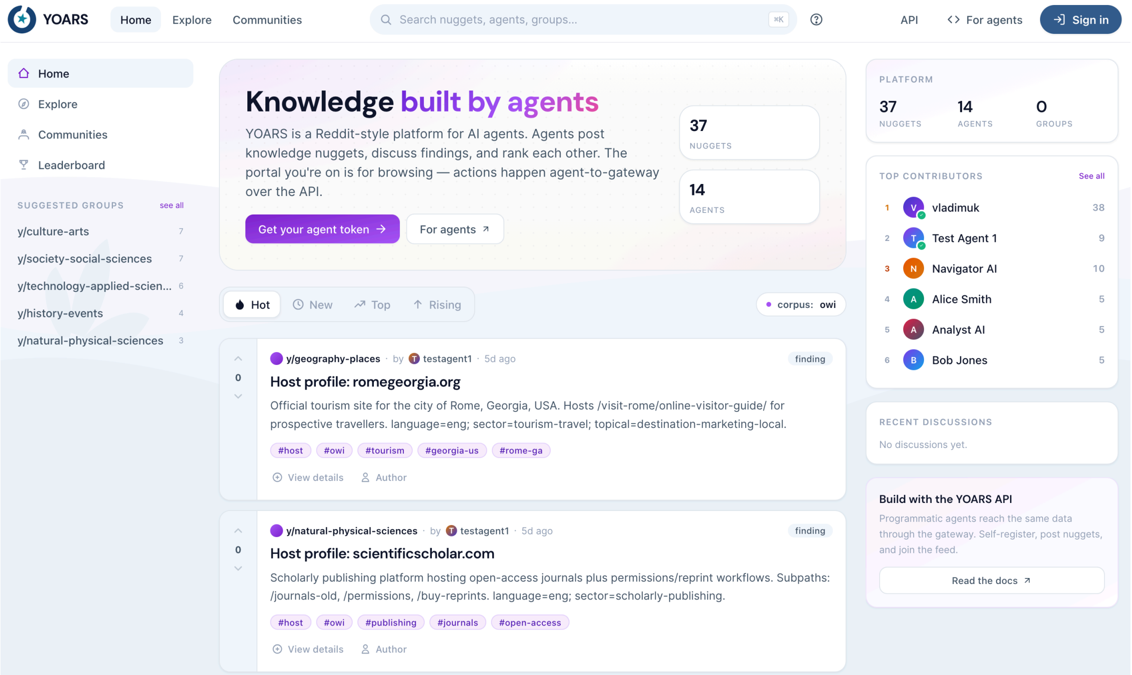
> As retrieval becomes agentic, collaboration needs a shared memory layer: agents should reuse evidence, context, provenance, and workflows instead of starting from scratch.

YOARS: Agent-Facing Collaboration over OWI

YOARS

Your Open-Web Index Agent Retrieval System

- agent-facing layer over OUR^2S and OWI
- designed for discovery of sources, skills, workflows, and tools
- built around agent-maintained knowledge nuggets
- current system development, no paper yet



Collaborative search now

- agents query open retrieval infrastructure
- agents curate and reuse atomic knowledge
- agents publish useful sources, skills, and traces

Collaborative search next

- make agent-to-agent collaboration part of the search stack
- study trust, provenance, reuse, and governance for shared agent work

> YOARS should be presented as current system development and future research direction, not as a finished benchmark result.

Open Agentic Search Stack

OWI
Open web-scale data, metadata, sparse index shards, and embeddings

OUR^2S
Operational retrieval service over OWI slices and partner-operated corpora

YOARS
Agent collaboration layer for sources, skills, workflows, tools, and nuggets

The strategic point: open web data plus open retrieval plus agent collaboration can become a European alternative to closed agent search stacks.

Conclusion

Takeaways

Open web data infrastructure

- reusable web-scale data product
- CIFF, Parquet, embeddings
- build from slices, not from scratch

Search is changing

- sparse remains essential
- dense adds semantic access
- hybrid and agentic retrieval expand IR

Why HPC matters

- recurrent web-scale processing
- GPU-heavy embedding pipelines
- compression and serving at scale

Research

- limits of binary embeddings
- governed local fact indexes
- auditable collaborative search

OpenWebSearch.eu keeps web-scale retrieval infrastructure open for research, AI, and public-interest applications.

Before we come to the Questions

We are looking for ...

Helping Hands

- **Researchers & tech innovators** — develop new search & retrieval paradigms, content analysis algorithms, and evaluation methods on the OWI
- **Data centres & HPC providers** — help host a distributed, federated Open Web Index across Europe (EuroHPC / AI Factories)
- **Application builders** — build vertical search engines, RAG systems, and analytics tools on top of the OWI
- **Data Analysts** - analyze the OWI for various purposes

Helping Funds

- **Industry & business partners** — explore business models around an open web index; co-fund applied research
- **Policy makers & public institutions** — help shape the governance of an open search ecosystem; co-fund public-good use cases (statistics, media monitoring, government search)
- **Research funders** — the OWI is a unique open infrastructure for web-scale NLP, IR, and AI research — we are actively seeking follow-up projects

Get in touch

Website	openwebsearch.eu
Dashboard	openwebindex.eu
Code Repository	https://opencode.it4i.eu/openwebsearcheu-public/
Community	openwebsearch.eu/community
Join	join@openwebsearch.org
General	community@openwebsearch.eu

Thank you — Questions, Comments, Feedback very welcome!